

Fine-Tuning XLM-RoBERTa for Latin POS Tagging: Challenges and Achievements

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1. Introduction

Part-of-Speech (POS) tagging is a fundamental task in Natural Language Processing (NLP) that plays a crucial role in downstream applications such as syntactic parsing and machine translation. The EvaLatin 2022 evaluation campaign (Sprugnoli et al., 2022) was designed to benchmark the performance of POS tagging systems for Latin, providing a competitive setting for testing different approaches.

The goal of this project is to develop a POS tagger capable of handling all POS subtasks in EvaLatin 2022. To achieve this, I fine-tuned XLM-RoBERTa base (Cunneau et al., 2020), a multilingual Transformer-based model, on the dataset provided by the campaign. The decision to use this model was guided by the campaign’s findings: the best-performing system leveraged XLM-RoBERTa large (Wróbel & Nowak, 2022) suggesting that multilingual pretraining benefits Latin POS tagging.

This study investigates whether XLM-RoBERTa base, the smaller model, can still achieve competitive performance, outperforming the baseline. The results demonstrate that, despite its smaller size, the model is highly effective for Latin POS tagging.

2. Background and Related Work

In recent years, interest in POS tagging for historical and ancient languages has increased,

driven by challenges such as CoNLL shared tasks (Zeman et al., 2017) and specialized competitions like the EvaLatin series (Sprugnoli et al., 2020, 2022). These initiatives have helped establish benchmarks and datasets for Latin, facilitating the development of state-of-the-art tagging models. The importance of these challenges and datasets for Latin NLP has been emphasized in a recent survey on machine learning for ancient languages (Sommerschield et al., 2023), which highlights how such resources have driven methodological advancements in the field.

The first EvaLatin challenge in 2020 saw participants using RNN-based architectures (Straka & Straková 2020; Wu & Nicolai 2020; Bacon 2020; Stoeckel et al. 2020) and gradient boosting techniques (Celano, 2020), with a focus on cross-genre and cross-time evaluation (Sprugnoli et al., 2020). Regarding the closed modality, the top-performing systems achieved an accuracy between 96.65% and 95.52% on the classical Latin subtask. However, their performance declined significantly when tested on cross-genre (ranging from 90.64% to 88.54%) and cross-time (87.00% to 83.96%) datasets. This drop highlights the challenges posed by diachronic and inter-genre variation, suggesting that even state-of-the-art models struggle with the linguistic diversity present in historical texts.

A significant breakthrough came with the adoption of Transformer-based architectures, as demonstrated in EvaLatin 2022 (Sprugnoli et al., 2022). The top-performing system (Wróbel & Nowak, 2022) fine-tuned XLM-RoBERTa, a

multilingual Transformer model, achieving state-of-the-art results in Latin POS tagging and morphological feature identification. Another strong submission (Mercelis & Keersmaekers, 2022) leveraged an ELECTRA-based model, supplemented by handcrafted rules for lemmatization.

In summary, the evolution of Latin POS tagging reflects a broader shift towards large-scale pre-trained language models. The EvaLatin 2022 results demonstrated that Transformer-based models, such as XLM-RoBERTa large, achieve state-of-the-art performance. This project builds upon these findings by fine-tuning XLM-RoBERTa base on the EvaLatin dataset, assessing whether a smaller Transformer model can still obtain good results.

3. Methodology

3.1 Dataset and pre-processing

The dataset used in this project is the EvaLatin 2022 corpus (Sprugnoli et al., 2022). The training set consists of 12 texts from five classical authors, totalling 316,573 tokens. While all texts are in prose, they represent various genres, including letters, treatises, and public speeches. The files of the texts are provided in CoNLL-U format.

Following the approach of Bacon (2020), an initial analysis was conducted to assess data quality and identify potential patterns. No missing labels were found. An examination of the POS tag distribution, shown in Figure 1, reveals three distinct frequency groups: (1) nouns and verbs, which are the most frequent; (2) NUM, X, and INTJ, which are the least frequent; and (3) all other tags, with frequencies ranging from approximately 25,000 (ADJ) to just over 8,000 (PART).

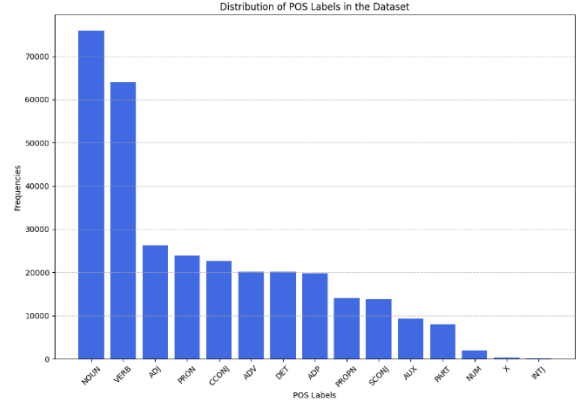


Figure 1. POS Tag Distribution in the training dataset.

The dataset split between the training and test sets was predefined by the campaign organizers. The test set comprises four distinct texts: (1) Book VIII of Livius' *Ab Urbe Condita* for the classical sub-task; (2) excerpts from Pliny the Elder's *Naturalis Historia* and Ovid's *Metamorphoses* for the cross-genre sub-task; and (3) *De Latinae Linguae Reparatione*, a work by the Renaissance humanist Sabellicus, for the cross-time sub-task. All test files are provided in CoNLL-U format without the correct labels, requiring participants' systems to automatically insert the predicted labels. These predictions were then evaluated by the campaign organizers using a Python script.

In this project, to assess the model's performance before the final test phase, I further split the training set into training and validation subsets, following a 92%–8% ratio. The validation texts were selected to ensure a balanced representation of different authors and genres. Table 2 and 3 presents the final division between the two sets.

AUTHORS	TEXTS	#TOKENS
Caesar	De Bello Gallico	44,818
Caesar	De Bello Civili (I, II)	17,287
Cicero	Philippicae (I–XIV)	52,563
Pliny the Younger	Epistulae (I–VIII, X)	60,695
Seneca	De Beneficiis	45,457
Seneca	De Clementia	8,172
Seneca	De Providentia	4,077
Tacitus	Historiae	51,420
Tacitus	Agricola	6,737
TOTAL	TEXTS	291,226

Table 1. Training data.

AUTHORS	TEXTS	#TOKENS
Cicero	In Catilinam	12,564
Seneca	De Vita Beata	7,270
Tacitus	Germania	5,513
TOTAL	TEXTS	25,347

Table 2. Validation data.

The tokenizer selected for this project is the XLM-RoBERTa tokenizer, which utilizes byte pair encoding (BPE) (Zouhar et al., 2023) to split already tokenized texts into sub-words when a given word is not found in the tokenizer’s vocabulary. This choice was driven by two main factors: (1) models generally perform better when paired with their corresponding tokenizer, and (2) BPE has proven effective in handling synthetic languages such as Latin (Parra, 2024).

Due to sub-word segmentation, the final pre-processing step involves aligning the labels with their corresponding sub-words. Specifically, each POS tag in the training set is assigned to the first sub-word of the corresponding original word, while a padding label is assigned to the remaining sub-words. However, the model pay attention to these sub-words; only the loss function is not computed for them.

3.2 Model

As previously mentioned, the selected model is XLM-RoBERTa base, a multilingual Transformer-based model that has been trained on a diverse set of languages, including Latin. Specifically, its pretraining corpus includes approximately 390 million Latin tokens (Conneau et al., 2020).

In this project, the model was evaluated on the validation set using different hyperparameter configurations, with accuracy calculated based on the key labels—those corresponding to the first sub-word of each original word. The best performance was achieved with the following hyperparameters, resulting in a training accuracy of 96.86% and a validation accuracy of 96.63%:

- batch size: 16,
- epochs: 5
- learning rate: 2e-5
- sequence length: 256
- gradient accumulation steps: 2

The experiments were conducted on Google Colab using the standard free T4 GPU. Due to computational constraints, optimization strategies such as gradient accumulation and mixed precision were employed to improve efficiency. The dataset was transferred via Google Drive, and to mitigate the risk of data loss during training, checkpointing mechanisms were implemented, allowing the model to be saved at regular intervals and training to be resumed if necessary.

3.3 Evaluation Metrics

The final evaluation was conducted using the Python script provided by the campaign organizers. This script ensures proper alignment of tokens, sentences, and words in the automatically labelled test set, preventing potential evaluation errors due to misalignment. Additionally, it computes key performance metrics, including precision, recall, F1 score, and accuracy.

4. Results and analysis

4.1 tests results

The model achieved an accuracy of 96.31% on the classical subtask test file. However, its performance declined to between 91.57% and 91.81% for the other subtasks. These results are broadly in line with those of other models from the 2022 evaluation campaign, as shown in Table 3.

Models	Classical	Cross-Genre	Cross-time
XLM-roBERTa Large	97.61	94.62 (0.22)	92.70
ELECTRA	96.33	92.31 (3.32)	92.11
XLM-roBERTa base	96.31	91.69 (0.001)	91.56

Table 3. The performances of the model compared to the others used in the campaign.

Examining the performance of models from the 2020 campaign (Sprugnoli et al, 2020), in which transformer-based architectures were not present, suggests that the accuracy gap between the classical and other subtasks was larger compared to XLM-RoBERTa base, especially with respect to the cross-time one. However, since the datasets differ, this hypothesis cannot be definitively confirmed. Nonetheless, the results indicate that even the base version of XLM-RoBERTa is capable of achieving strong performance.

4.2 Error analysis

The confusion matrix show in figure highlights several patterns in the model's misclassifications. As noted by Wróbel and Nowak (2002), the most significant errors made by XLM-RoBERTa models in Latin involve nominal categories. Specifically, adjectives (ADJ), nouns (NOUN and PROPN), and verbs are frequently misclassified, likely due to their morphological similarity in Latin.

Overall, these errors suggest that while the model performs well, it faces challenges in distinguishing between categories with subtle morphological or syntactic differences, a common issue in Latin POS tagging.

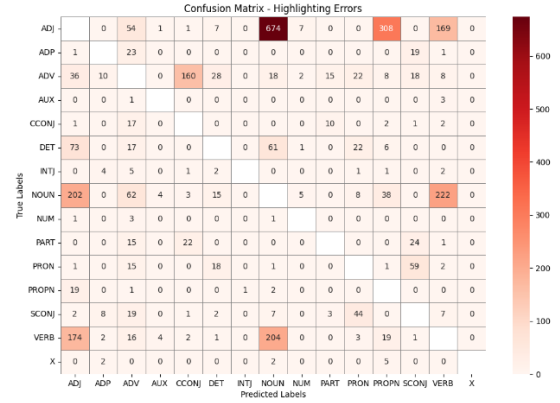


Figure 2. Confusion matrix.

5. Conclusions

This study aimed to evaluate whether the base XLM-RoBERTa model could achieve strong results in POS tagging for Ancient Latin. Despite slightly lower performance than the best systems in the 2022 campaign and the inability to directly compare results with the 2020 edition, the model still demonstrated high accuracy. These findings suggest that even the base version of XLM-RoBERTa is effective for this task, reinforcing its potential for Latin NLP applications.

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